

Sentiment Analysis of the Depp V. Heard Trial (2022)

Introduction/Background

In 2022, Johnny Depp sued his former wife Amber Heard, for defamation over an Op-Ed she had written two years prior in which she claimed to have been a victim of domestic abuse. The case received a large amount of attention from the public due to the fact that the trial was livestreamed in real time. This trial was the second part to a first trial that happened in 2020 in which Amber Heard's accusations of domestic abuse had proven to be true, severely damaging Depp's reputation. Over the course of the trial (from April 2022 to June 2022), on Tik Tok alone, the hashtags related to the case had 20 billion views (Espada, 2023). As the trial trended on social media and new information was revealed, the previous public opinion of Depp being an abuser changed to a new public opinion that Depp was the victim of abuse.

This case was so televised that even the streaming platform, Netflix, released a documentary series in which they examine the effects of social media on the trial. The document talks about how public opinion perhaps had a huge impact on the jury when it came to giving a verdict, since the public overtime became overwhelmingly skewed towards Depp's side. Whether social media biased the jury's decision is hard to determine, but what remains clear is the power of social media in allowing a grand scale of online engagement and the power it holds in shifting sentiment within a short period of time.

Research Question and Goal

This study addresses the following research questions: How does public sentiment on X (Twitter) evolve during online controversies or scandals? Specifically, how did sentiment toward Johnny Depp in contrast to that of Amber Heard's evolve during the 2022 trial, and what

temporal patterns in sentiment polarity and intensity can be derived from the analysis of textual data from highly engaged tweets?

This study aims to illustrate how digital platforms mediate and amplify public sentiment in real time, particularly during highly publicized controversies, using dictionary-based sentiment analysis combined with temporal aggregation.

Literature Review

With the rapid development of technology in recent years, we can observe the parallel development of social media. Just as of 2023, there were around five billion social media users worldwide, and this trend aims to increase to over 6.6 billion by 2030 (Statista, 2025). A growing body of research suggests that people coming in contact with other people's opinions on social media can influence others' own opinions and their willingness to participate in social media discussions (Neubaum & Krämer, 2017). Recent research has aimed to explore and help understand what are the mechanisms behind this influence phenomenon. Some suggest that it could be the role of social media in facilitating the emergence of echo chambers and opinion polarization (Bauman et al., 2020), people's fear of isolation (Neubaum & Krämer, 2017), and its capacity to enable a break from Noelle-Neumann's theory of the spiral of silence (Laor, 2024).

Furthermore, the influence of media coverage on public opinion has been particularly evident in the specific context of high-profile celebrity trials. A notable precedent is the 1995 OJ Simpson trial, which was one of the first criminal trials to be broadcasted live on television, drawing massive public attention and sparking widespread debate about the relationship between media exposure and public perception of legal proceedings (Brown et al., 1997). Much like Depp, Simpson was a beloved and iconic public figure, which made the public's emotional investment in the trial's outcome particularly intense. However, what fundamentally

distinguishes two trials is the role of social media. While the Simpson trial unfolded through the lens of traditional broadcast media, the Depp vs. Heard trial played out in real time across platforms like Twitter and TikTok, where the audience was an active participant sharing its opinions, creating content, and collectively shaping the narrative in real time. This distinction is critical, as it suggests that social media does not merely transmit public opinion but actively constructs it.

To study shifts in public opinion at scale, this project employs computational sentiment analysis, an approach increasingly used in social media research to capture real-time public discourse. Two lexicon-based methods are employed: 1) the AFINN lexicon, which assigns numerical sentiment scores to individual words on a scale from -5 to +5, and was specifically constructed and validated for sentiment detection in microblog contexts such as Twitter (Nielsen, 2011); 2) the NRC emotion lexicon, to capture discrete emotional dimensions such as anger, fear, joy, and trust, that was generated through large-scale crowdsourcing to produce a high-quality word-emotion association resource (Mohammad & Turney, 2013). Together, these tools allow for both polarity tracking and nuanced emotional analysis over time, making them well suited for event-driven analysis of the kind undertaken here.

Despite its strengths, dictionary-based sentiment analysis carries known limitations. Some evidence suggests that sarcasm alone could account for as much as a 50% drop in accuracy when automatically detecting sentiment (Sykora et al., 2020), as lexicon-based tools struggle to interpret ironic or context-dependent language. Furthermore, tweet-level sampling ignores that some users are significantly more active than others, potentially biasing samples toward the most vocal participants rather than representing the broader population (Peled-Raz et al., 2023). This study addresses these limitations through a deliberately small, manually verified sample of

high-engagement tweets. While this restricts generalizability, it prioritizes analytical accuracy over scale which we believe is a reasonable tradeoff for this event-driven, exploratory study.

Data and Methods

Initially, data was planned to be collected independently through Twitter scraping. Unfortunately, such an approach proved to be unreliable for historical analysis. Advanced search for 2022 tweets did not consistently return the most relevant or highest-engagement content, and repeated queries produced inconsistent results. The lack of reproducibility, therefore, made systematic data collection impractical. Instead, this project uses publicly available Twitter (X) data from the Johnny Depp and Amber Heard Twitter Data dataset on Kaggle, specifically the `fulltrialdata.csv` file compiled by user Luke Groves. The dataset contains 18,323 English-language tweets across 90 variables, spanning April 27 to May 13, 2022. This 17-day period is considered a period of active courtroom testimony with high public engagement from social media platform users.

Data preprocessing was conducted in R. Tweet text was converted to lowercase and cleaned by removing URLs, HTML entities, punctuation, non-alphabetic characters, and extra whitespace. Unnecessary variables were dropped, and the dataset was filtered to retain only tweets referencing Johnny Depp or Amber Heard. Each remaining tweet was then categorized into one of three groups based on keyword presence: `depp` (only Depp mentioned), `amber` (only Heard mentioned), and `mix` (both individuals mentioned). As shown in Fig. 1, the overwhelming majority of trial-era tweets mention both figures together, so the `mix` group dominates raw volume while single-mention tweets are comparatively rare. Because the central research question is comparative, the subsequent vocabulary, emotion, and sentiment analyses contrast the `depp` and `amber` groups directly.

Text was then tokenized into individual words and lemmatized to reduce words to their base forms. Standard English stopwords were removed, along with a custom set of domain-specific stopwords (including the principals' names, platform-related terms, and generic trial vocabulary), so the analysis would focus on substantively meaningful language while preserving the original tone of the tweets.

A central methodological concern is that a small number of highly active accounts can dominate tweet-level samples and bias results toward the most vocal participants. To mitigate this, the sentiment and emotion measures reported here are user-weighted: contributions are aggregated so that prolific posters do not exert disproportionate influence on the aggregate signal.

Two lexicon-based methods were implemented for sentiment and emotion. First, the AFINN lexicon was used to assign numerical sentiment scores (ranging from -5 to +5) to individual words, and a user-weighted average score was computed per group and date to track relative sentiment over time. Because trial vocabulary is intrinsically negative ("abuse," "violence," "victim," "defamation"), AFINN values sit below zero for both groups; the meaningful quantity is therefore the depp-versus-amber contrast on a given day rather than the absolute sign. Second, the NRC emotion lexicon was used to characterize the eight discrete emotions it encodes — anger, fear, trust, sadness, anticipation, disgust, joy, and surprise. For each group, the user-weighted proportion of emotion-labeled words falling into each category was computed, yielding an overall emotion profile that can be compared across the two groups.

To explore dominant themes, word-frequency analysis was conducted separately for each group. The most frequent terms were rendered as word clouds, and the top words per group were also examined as normalized proportions (rather than raw counts) and broken out by month, so

that narrative shifts between April and May could be observed without the larger group appearing to use every word more often. In addition, term frequency–inverse document frequency (TF-IDF) was used to identify the words most distinctive to each group — terms that characterize one group’s discourse relative to the other. All analyses were visualized using the ggplot2 package (with ggwordcloud for the word clouds).

Findings

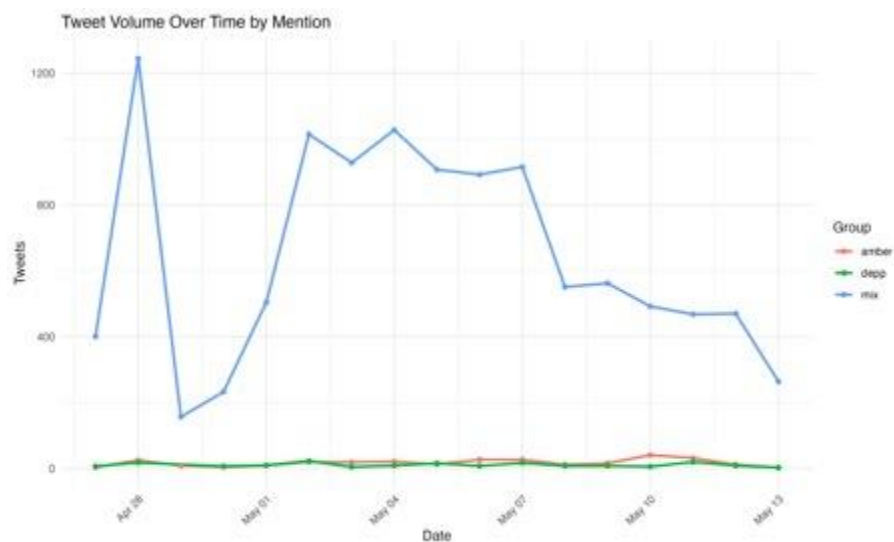


Fig. 1. Tweet volume over time by mention group.

The volume of trial-related tweets is overwhelmingly driven by posts that mention both Depp and Heard. The mix series peaks sharply around April 28 (over 1,200 tweets) and again in early May (roughly 1,000), then declines steadily through May 13. By contrast, tweets mentioning only one party remain low throughout the entire window. This pattern indicates that the online conversation largely treated the two figures as a single, intertwined story rather than discussing either in isolation.

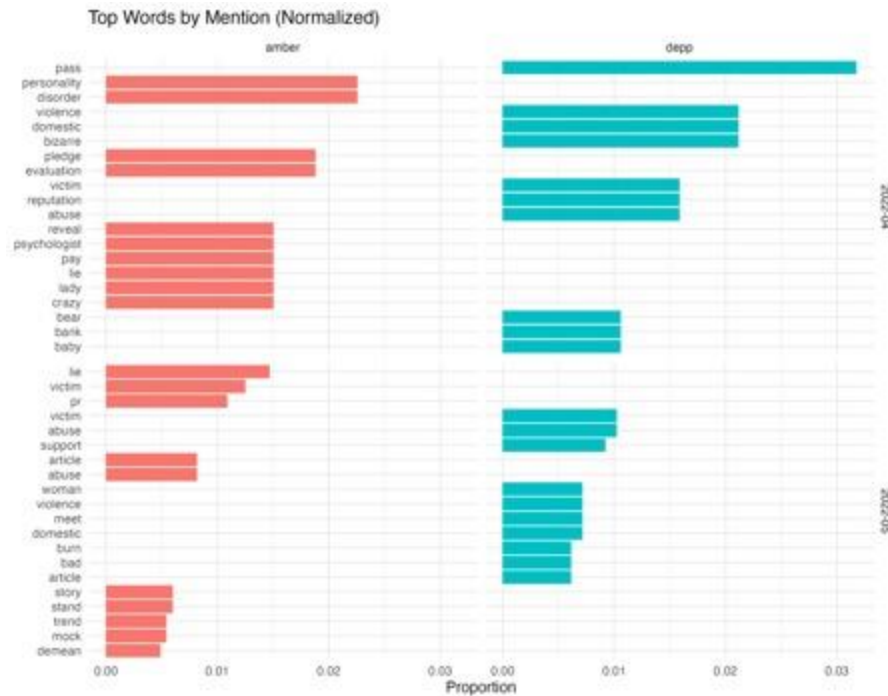


Fig. 3. Top normalized words by mention group and month.

The normalized, month-by-month word frequencies show how framing shifted over the course of the trial. In April, amber-mention discourse centered on personality, disorder, evaluation, psychologist, and pledge — language tied to psychological testimony and the disputed charitable-donation pledge — while depp-mention discourse foregrounded violence, domestic, reputation, and victim. By May, amber-mention tweets shifted toward lie, pr, mock, and demean, and depp-mention tweets toward support, victim, and woman, indicating that the conversation moved from case specifics toward more evaluative and emotionally charged language on both sides.

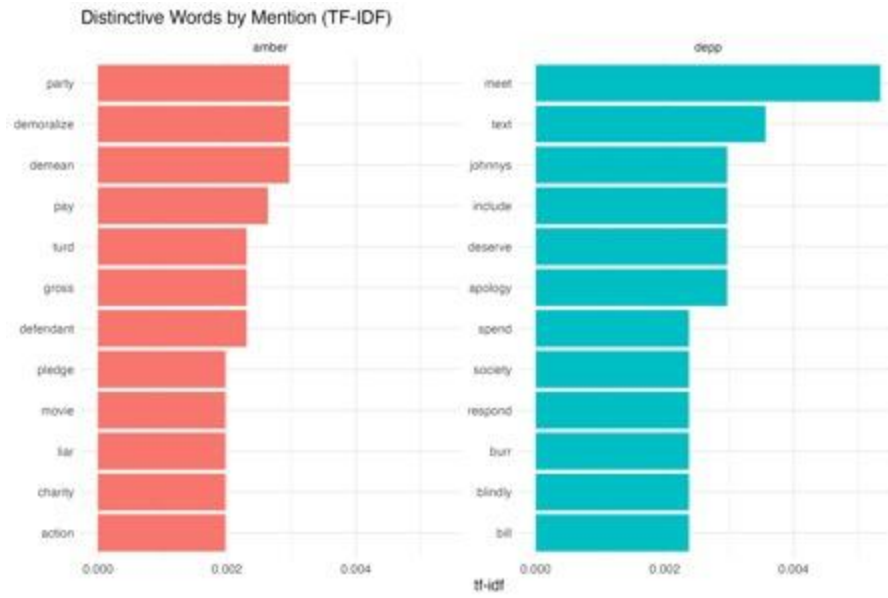


Fig. 4. Distinctive words by mention group (TF-IDF).

The TF-IDF analysis isolates each group's most characteristic vocabulary. Amber-distinctive terms include party, demoralize, demean, pay, turd, gross, defendant, pledge, and charity — several of which map onto widely circulated moments of testimony and the donation-pledge dispute. Depp-distinctive terms include meet, text, apology, deserve, include, spend, and society, reflecting attention to entered evidence (such as text messages) and to questions of accountability and reputation.

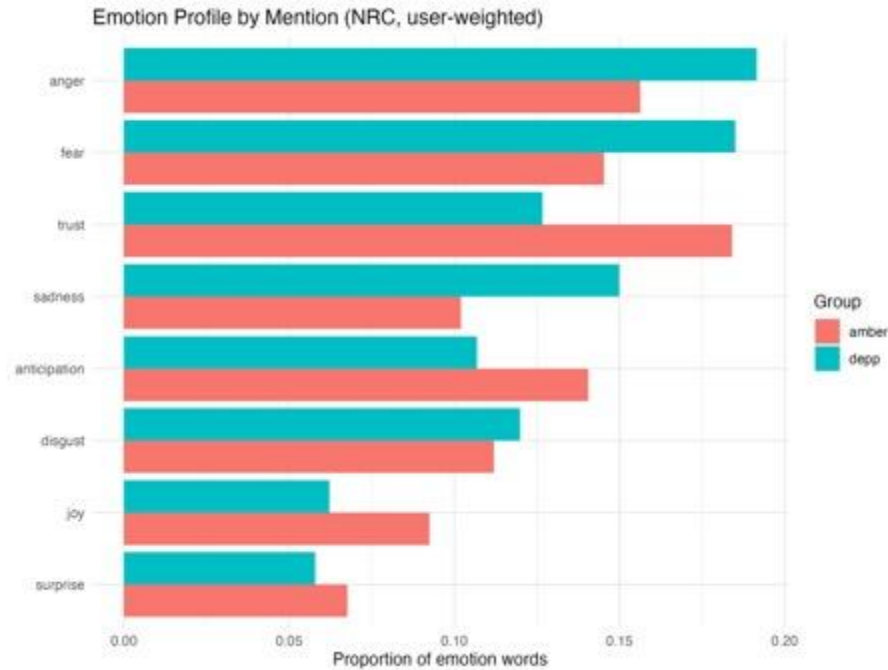


Fig. 5. Emotion profile by mention group (NRC, user-weighted).

The emotion profiles differ in shape between the two groups. Tweets mentioning Depp register higher proportions of anger, fear, sadness, and disgust, whereas tweets mentioning Amber carry comparatively more trust, anticipation, joy, and surprise. This contrast should be read with care: grouping is based on who is mentioned, not on the author's stance, so emotionally intense tweets that criticize one party are counted under that party's mention. Lexicon-based emotion scoring also cannot resolve sarcasm or negation, both of which are pervasive in this discourse.

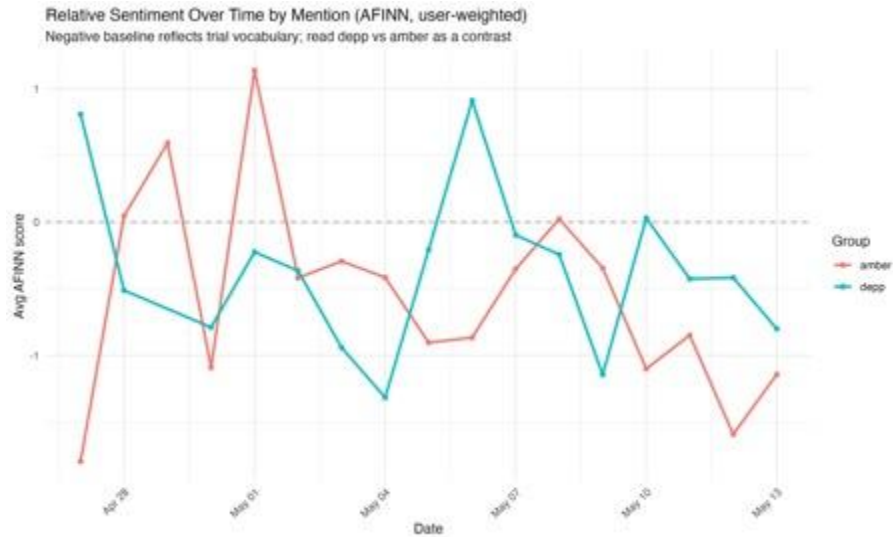


Fig. 6. Relative sentiment over time by mention group (AFINN, user-weighted).

The daily sentiment series are anchored in negative territory for both groups — an expected consequence of the trial’s vocabulary — and are highly volatile, with the depp and amber lines repeatedly crossing rather than separating cleanly. Both groups show sharp, short-lived swings (for example, an amber-mention spike around May 1 and depp-mention peaks at the start of the window and in early May), but neither series sits consistently above the other, and both end the period in negative territory. The dictionary-based polarity measure thus captures day-to-day reactivity and churn more than a smooth, monotonic divergence between the two groups.

Discussion

Across the 17-day window, the analysis points to a conversation that was intensely interconnected and lexically distinct between groups, but more ambiguous in its raw sentiment polarity than a simple “pro-Depp versus anti-Heard” narrative would suggest. The volume data show that most engagement referenced both parties simultaneously, consistent with a single shared media event rather than two separate conversations. Within that event, the vocabulary

analyses (Figs. 2–4) reveal genuinely different framings: amber-mention discourse gravitates toward the mechanics of the defamation case (credibility, psychological evaluation and the donation pledge), while depp-mention discourse leans on support, reputation, and victimhood language. These differences are stable enough to surface consistently across the word clouds, the normalized frequencies, and the TF-IDF rankings.

The emotion and sentiment results are more nuanced. The NRC profile (Fig. 5) shows depp-mention tweets skewing toward anger, fear, and disgust and amber-mention tweets toward trust and anticipation — a pattern that most likely reflects the mention-based grouping rather than endorsement, since outrage expressed on Depp’s behalf is counted within the depp group. The AFINN series (Fig. 6) reinforces this caution: rather than a clean, widening gap, the two groups’ daily sentiment scores are volatile and frequently cross, both held down by the negativity inherent in trial vocabulary. Periods of heightened activity in early May coincide with widely covered testimony, including arguments on May 2 that Depp lost major career opportunities after the allegations became public, and clinical testimony on May 3 regarding Heard, which may help contextualize some of the short-term swings, though the dictionary-based measures alone cannot establish that link.

Taken together, these limitations temper how far the sentiment results can be pushed: lexicon methods cannot reliably handle sarcasm and negation, mention-based grouping is only a coarse proxy for stance, and the observation window is short. User-weighting reduces, but does not eliminate, the influence of the most active accounts. For future work, the window could be extended beyond early May and across the full trial; bigrams and context-aware NLP models could be introduced to better capture negation, sarcasm, and coordinated behavior; and the

results could be compared against other platforms such as Reddit or TikTok to examine whether the same framing and emotional patterns hold.

Conclusion

This study examined how online discourse surrounding the 2022 Depp v. Heard trial was structured across volume, vocabulary, emotion, and sentiment. The conversation was dominated by tweets engaging both figures at once, and the two mention groups used clearly distinguishable vocabularies — amber-mention discourse oriented around the defamation case, credibility, and psychological evaluation, and depp-mention discourse around support, reputation, and victimhood. Emotionally, depp-mention tweets carried more anger, fear, and disgust, while amber-mention tweets carried more trust and anticipation, though these differences reflect who was mentioned rather than the authors' stances. Crucially, dictionary-based polarity did not reveal a clean, monotonic divergence between the groups: both registers remained negative and volatile throughout the period. These findings illustrate both the value and the limits of lexicon-based sentiment analysis for event-driven social media discourse, and they motivate future work using context-aware methods, longer observation windows, and cross-platform comparison.

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